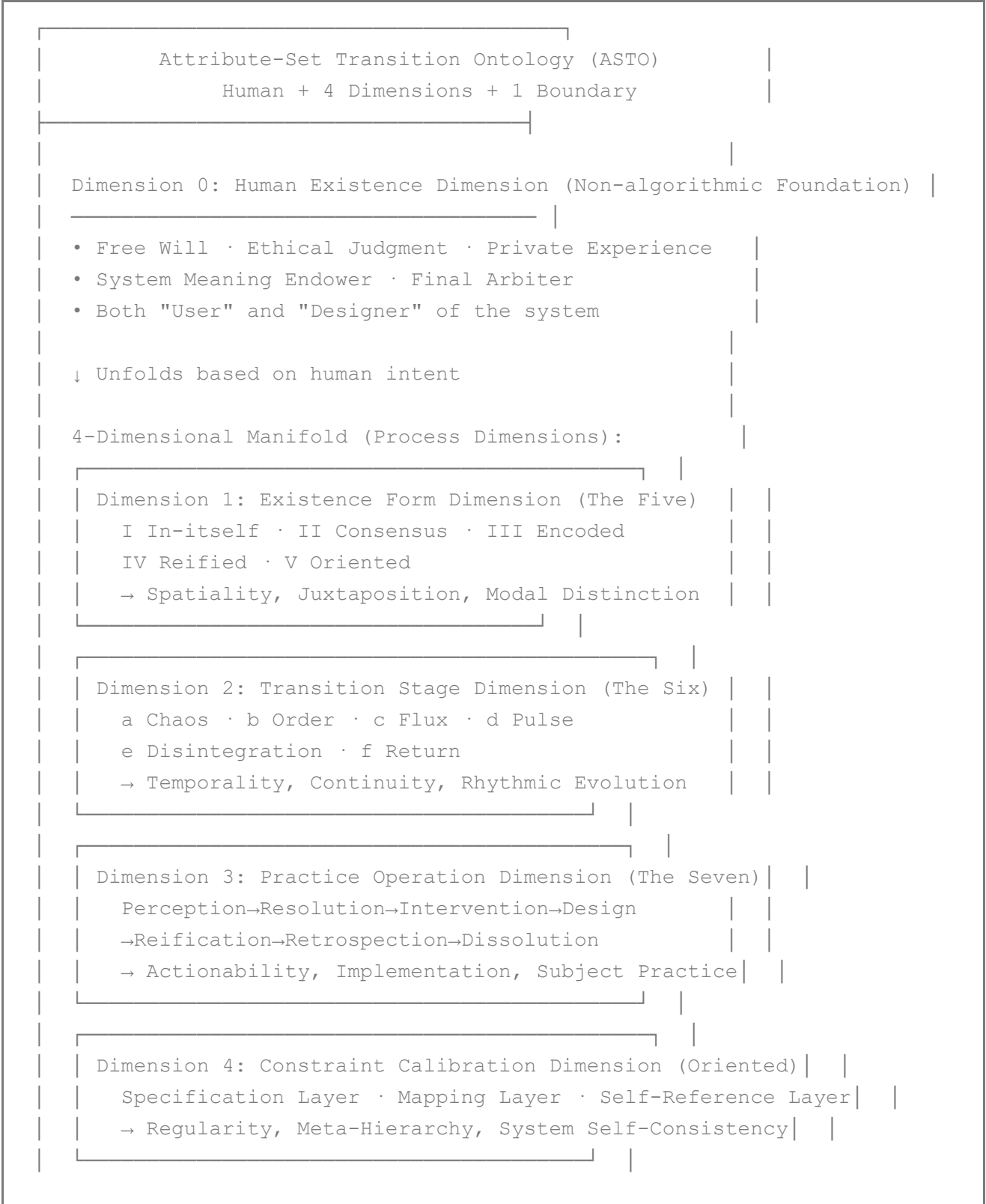


ASTO Complete Practice Loop System: Core Dynamics Mechanism

I. System Panorama: Self-Consistent Structure of Human + 4 Dimensions + 1 Boundary



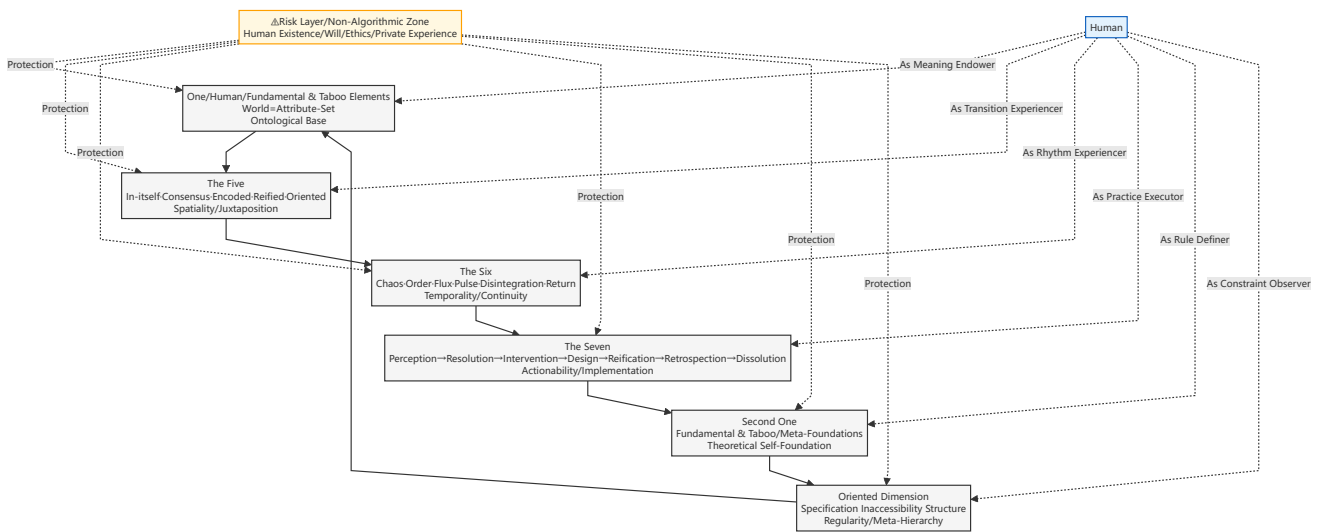
↓ Results feed back to Human

+1 Boundary: Risk Layer (Protection Dimension)

- Prevent 4D manifold from swallowing Dimension 0
- Protect human subjectivity from algorithmic dissolution
- Ensure irreducibility of freedom, ethics, experience

II. Core Dynamics: 1→5→6→7→1 Practice Loop

Loop Unfolding Details



Stage Function Analysis

Stage	Function	Corresponding Axiom/Theorem	Output
One	Ontological Foundation · Definition of Fundamental & Taboo Elements	Axiom -1 (Language Limitation) Axiom 0 (Environmental Entropy Increase)	Meta-definition of Attribute-Set
The Five	Existence Modal Unfolding · State Transition	Axiom 1 (Structural Homeostasis) Axiom 2 (Attribute Stratification)	Current Existence Pattern

Stage	Function	Corresponding Axiom/Theorem	Output
The Six	Transition Rhythm Evolution · Time Dynamics	Axiom 0 (Environmental Entropy Increase) Axiom 3 (Compliance Transmission)	Change Stage/Intensity
The Seven	Subject Practice Operation · Intervention Loop	Axiom 4 (Motility Field) Axiom 8 (Practice Loop)	Concrete Action Result
Oriented	Rule Constraint Calibration · System Self-Consistency	Theorem 4 (Boundary is Freedom) Axiom 11 (Freedom)	Corrected Rule Set
Risk Layer	Subject Protection · Non-Algorithmic Retention	Axiom 7 (Position of Human) Axiom 8 · Lemma 8.1 (Cognitive Asymmetry)	Safe Boundary

III. Three Pairs of Core Tensions: Power Source of System Evolution

1. Structure vs Agency Tension

- **Structural Dimension:** The Five (Existence Pattern) + The Six (Transition Rhythm) + Oriented Dimension (Specification Constraint)
- **Agency Dimension:** The Seven (Subject Practice) + Human Free Will
- **Tension Manifestation:** Confrontation and synergy between structural trends and subject agency
- **Dynamic Role:** Propels system balance between stability and change

2. Order vs Chaos Tension

- **Order End:** b Order Stage + III Encoded State + Specification Layer
- **Chaos End:** a Chaos Stage + I In-itself State + e Disintegration Stage
- **Tension Manifestation:** Dialectical relationship between organizational needs and creative destruction
- **Dynamic Role:** Drives system cycling between organizational optimization and innovative breakthrough

3. Algorithm vs Ethics Tension

- **Algorithm End:** IV Reified State + Seven Orders Automation + Mapping Layer
- **Ethics End:** V Oriented State + Risk Layer + Human Adjudication
- **Tension Manifestation:** Fundamental conflict between efficiency optimization and value judgment
- **Dynamic Role:** Ensures technical rationality does not swallow humanistic care

IV. Three-Layer Nested Feedback Loops: Anti-Degradation Mechanism

Operation Layer Feedback (The Seven Loop)

Perception → Resolution → Intervention → Design → Reification →
Retrospection → Dissolution



- **Frequency:** Highest (Millisecond–Minute level)
- **Function:** Rapid adjustment of specific actions
- **Anti-Degradation:** Prevent operational ossification through continuous retrospection and dissolution

Rule Layer Feedback (Oriented Dimension Self-Reference)

Specification Layer (Explicit Ban) → Mapping Layer (State Forbidden Zone)
→ Self-Reference Layer (Specification Self-Check)



- **Frequency:** Medium (Hour–Day level)
- **Function:** Dynamic adjustment of rule constraints
- **Anti-Degradation:** Prevent rule obsolescence through self-reference layer paradox detection

Ontological Layer Feedback (Loop Closure)

One (Attribute-Set) → 4D Manifold (Process Unfolding) → Oriented
Dimension (Rule Constraint)



- **Frequency:** Lowest (Month–Year level)
- **Function:** Update theoretical foundation

- **Anti-Degradation:** Prevent theory from detaching from practice through complete loop verification

V. Three Roles of Human in the System

1. Internal System Basis (Not External User)

- **Meaning Endower:** Without human intentionality, the system is just an empty structure
- **Value Anchor Point:** Non-algorithmic dimensions like ethical judgment, aesthetic experience
- **Final Arbiter:** When algorithms encounter "undecidable" boundary cases

2. Transition Helmsman

- **The Seven Executor:** Proactively guide transition through Perception-Resolution-Intervention...
- **Rhythm Tuner:** Adopt different strategies at different stages of The Six
- **Mode Switcher:** Migrate between The Five states, changing the way of existence

3. Risk Guardian

- **Guard of Non-Algorithmic Zone:** Protect will, ethics, and experience from dissolution
- **Motility Right Allocator:** Decide who has the right to change the system and what to change
- **Obsolescence Justice Arbiter:** Judge what should be eliminated and what should be preserved

VI. Practice Guide: How to Use This Framework

When Analyzing Any System Transition

1. Locate Current Status:

- In which mode of The Five? (Existence Form Dimension)
- In which stage of The Six? (Transition Stage Dimension)

2. Identify Operation Mode:

- Which step of The Seven is being executed? (Practice Operation Dimension)
- What rules constrain it? (Constraint Calibration Dimension)

3. Assess Human Role:

- What is the human's role in the system? (Dimension 0)
- Which aspects are protected by the Risk Layer? (+1 Boundary)

4. **Predict Evolution Direction:**

- How are the three pairs of tensions currently acting?
- How do the three layers of feedback prevent degradation?

When Designing Autonomous Systems

1. **Ensure Human Reservation Zone:** Clearly define boundaries between algorithmic processing and mandatory human adjudication
2. **Establish Complete Feedback:** Build three-layer feedback of operation, rule, and ontology
3. **Balance Core Tensions:** Intentionally introduce tensions of structure vs agency, order vs chaos in design
4. **Achieve Loop Closure:** Ensure practical results can feed back to correct theoretical foundations

VII. ASTO Dynamics Core Formulas